

DATA COMMUNICATION

- ▶ Data communication refers to the exchange of data between two or more networked or connected devices.
- ▶ **data communication** is communication in which we can send or receive data from one device to another

DATA COMMUNICATION COMPONENT

A data communications system has five components

1. Message.

Message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. Sender

Sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver

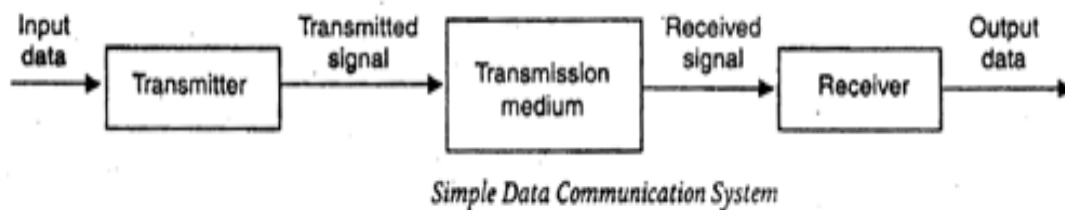
Receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. Transmission medium

Transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

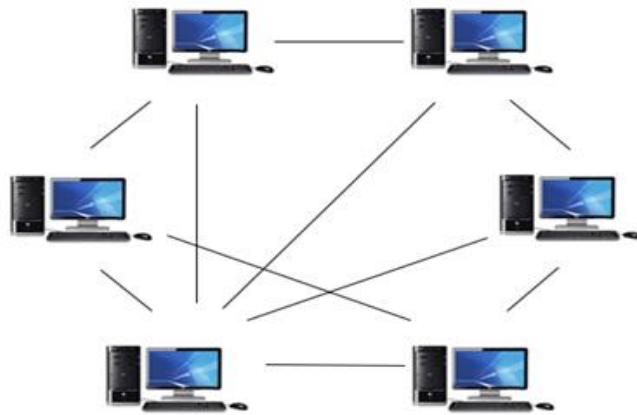
5. Protocol

A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.



COMPUTER NETWORK

A computer network is a set of devices connected through links. A node can be computer, printer, or any other device capable of sending or receiving the data. The links connecting the nodes are known as communication channels.



USE/NEED OF COMPUTER NETWORK

- To Share Computer Files
- To share Hardware Device
- Application sharing
- Communication
- Gaming
- Voice over IP
- Resource sharing
- Internet access
- E-Commerce
- Improve Communication Speed & Accuracy
- To reduce the cost of data transfer

Applications of Computer Networks

1.Communication Medium

The most widely used network application is its use as a communication medium. It is possible for everyone to send and receive e-mails.

Sharing: A Major applications of network is that it facilitates distinct types of sharing:

- Sharing of Peripherals
- Sharing of Data and Information

2.Access to remote database

Another area of network application is access to remote database. Airline reservation, hotel booking, railway reservation etc. are easy for anyone sitting anywhere in the world.

3.Financial Services

It depending on computer networks include foreign exchange and investment services, electronic funds transactions (EFT) etc.

4.Information Services

There may be corporate information systems such as marketing information system, customer information system, product information system etc. Computer networks also facilitate decision making.

5.Tele Conferencing

Tele Conferencing allows conferences to occur without the participants being in the same place.

6.Entertainment

It applications include video on demand interactive films and games, live television where audience participate in quiz shows etc.

7.Marketing and Sales

Computer networks are used extensively in both marketing and sales organizations. Marketing professionals use them to collect, exchange and analyze data relating to customer needs.

8.Business Applications

Now a day, business affairs rely on computer networks and internet works. To monitor production, inventories, to make payments etc. computers are extensively in use.

Advantages of Computer Networks

1.Resource Sharing

The networking of computers allows to share resources with others on the network. This permits to share software and peripherals such as printers, plotters, scanners and so on.

2.Easier Backups

It means making copy of data for future use. In Business or office work, backup data is very important and valuable.

3.Communication

Organizations today are widely dispersed over in the world. They need to exchange information and data. The computer network allows easy communication through email, internet telephony,

teleconferencing etc. This allows the communication between various organizations to cooperate with each other in performing certain task rapidly.

4.Interactive Entertainment

Computer network provides many types of games and entertainment. The real time interactive games like flight simulators are also becoming popular.

5.Cost Reduction

Sharing of hardware and software resources leads to the reduced equipment costs. Also cost of transfer of data/documents over long distances using computers connected on network is much cheaper than other means of data transfer like speed cost, telegram etc

Disadvantages of Computer Networks

1.Data Security: Unauthorized Personnel can access the information if network security is weakly implemented.

2.Expensive: They are expensive to purchase and maintain.

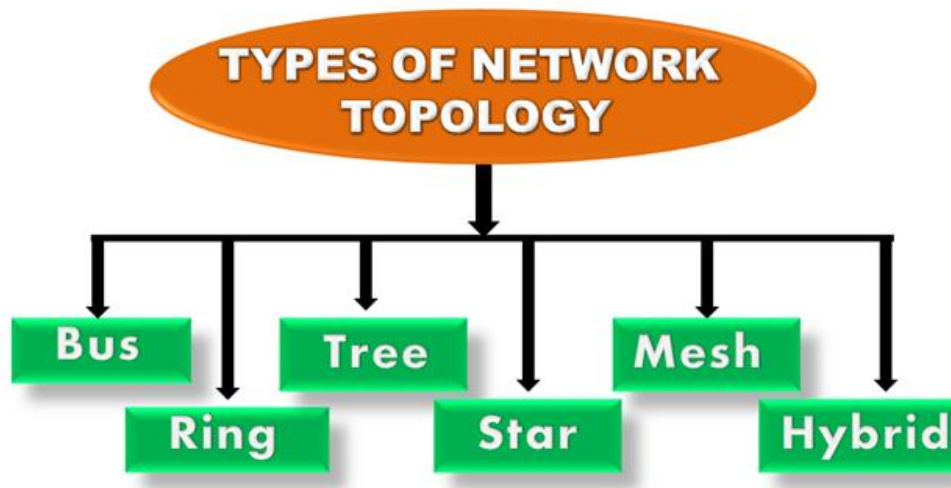
3.Crashes: On a server-based network, if the server crashes, work gets disrupted. If proper precautions are not taken to ensure regular backups, the crash may result in the loss of critical data.

4.Privacy: If the network is poorly implemented, improper communication may take place. This may lead to a situation in which the private or important e-mails can be read by other users and this result in loss of privacy.

5.Viruses: The virus on one system in a network can affect the whole computer network. Viruses are easily spread over networks. To overcome this problem costs more time, money and administrative effort.

NETWORK TOPOLOGY

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection. It is the structure of network that tell how devices are connected with each other.



1.BUS Topology

Bus topology is a network type in which every computer and network device is connected to single cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.

Features of Bus Topology

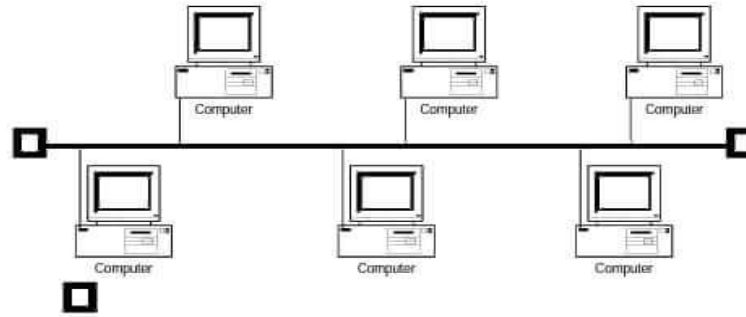
- ❖ It transmits data only in one direction.
- ❖ Every device is connected to a single cable

Advantages of Bus Topology

1. It is cost effective.
2. Cable required is least compared to other network topology.
3. Used in small networks.
4. It is easy to understand.
5. Easy to expand joining two cables together.

Disadvantages of Bus Topology

1. Cables fails then whole network fails.
2. If network traffic is heavy or nodes are more the performance of the network decreases.
3. Cable has a limited length.
4. It is slower than the ring topology.



2. RING Topology

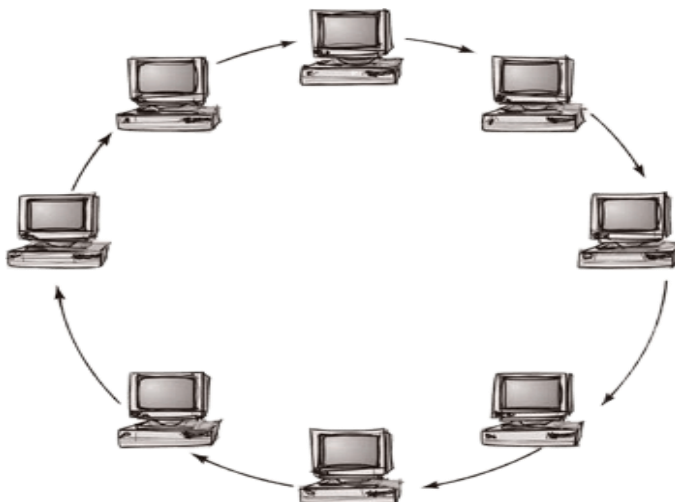
It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbours for each device.

Advantages of Ring Topology

1. Transmit the data in high speed
2. It easy to manage
3. We can Expand the topology
4. It can be easy to setup

Disadvantage of ring topology

1. Adding or deleting the computers disturbs the network activity.
2. Failure of one computer disturbs the whole network.
3. Speed slow after adding new computer
4. Damage the wire disturb the whole network



3.STAR Topology

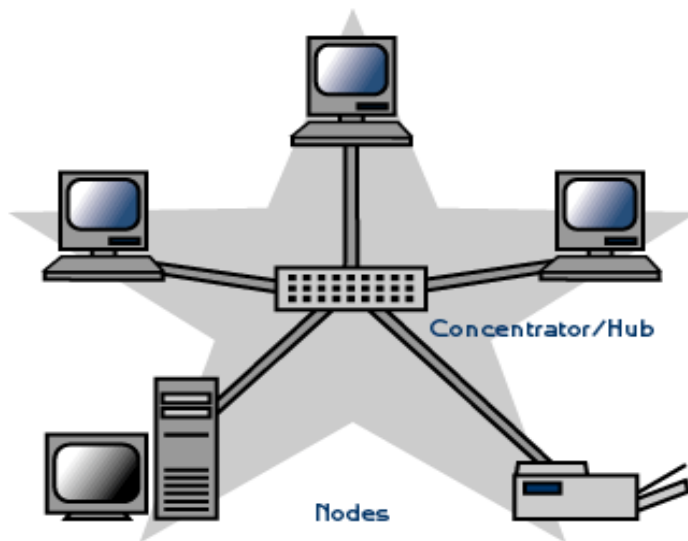
In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node.

Advantages of Star Topology

1. Fast performance with few nodes and low network traffic.
2. Hub can be upgraded easily.
3. Easy to troubleshoot.
4. Easy to setup and modify.
5. Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

1. Cost of installation is high.
2. Expensive to use.
3. If the hub fails then the whole network is stopped because all the nodes depend on the hub.
4. Performance is based on the hub that is it depends on its capacity



4.Mesh topology

- ❖ A mesh topology is a type of computer network in which each node (computer or other device) is connected to every other node in the network.

- ❖ In mesh topology each device can communicate directly with each other.

Features of Mesh Topology

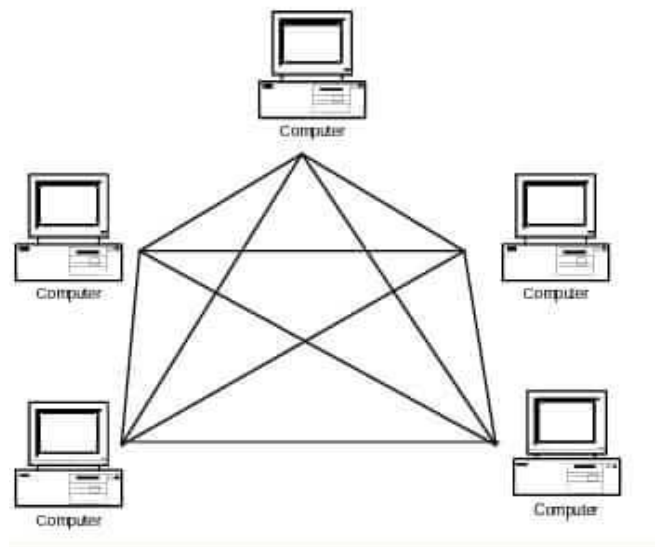
- Fully connected.
- Robust.
- flexible.

Advantages of Mesh Topology

- ❖ Each connection can carry its own data load.
- ❖ It is robust.
- ❖ Fault is diagnosed easily.
- ❖ Provides security and privacy.

Disadvantages of Mesh Topology

- ❖ Installation and configuration is difficult.
- ❖ Cabling cost is more.
- ❖ Bulk wiring is required



5.HYBRIDTopology

It is two different types of topologies which is a mixture of two or more topologies. For example, if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

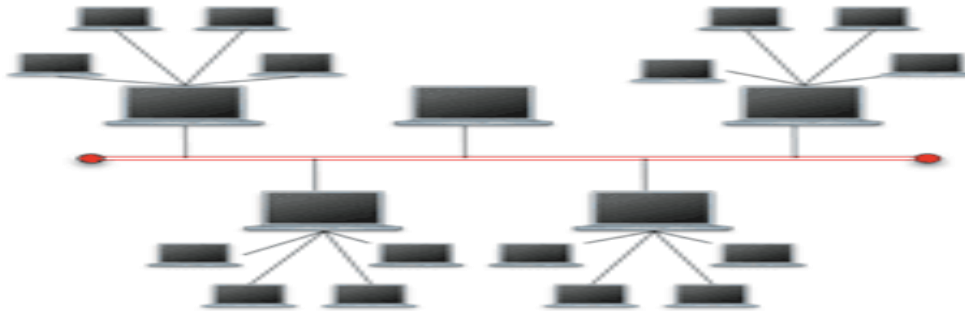
Advantages of Hybrid Topology

Reliable as Error detecting and troubleshooting is easy.
It will be Effective.

Scalable as size can be increased easily.
Hybrid Topology is Flexible.

Disadvantages of Hybrid Topology

Complex in design.
It can be Costly.



Types of Computer Networks:

- a. **LAN (Local Area Network):** It refers to two or more up to several hundred computers and variety of computing equipment, linked via some communication lines, to exchange information speedily, within a small well-defined area. For Example: University, Campus etc.
 - **Characteristics of LANs:**
 - Small Area
 - Single Owner
 - Reliability and Stability
 - Physically Secured
 - Flexibility
 - Low cost data transmission
 - Data and Resource Sharing
 - **Advantages of LANs:**
 - LAN Can be used for Sharing Resources
 - We can exchange the data and file in high speed.
 - Data Transaction
 - LAN Provide the Centralized Backup
 - Easy and Economical Communication
 - LAN provide the Good Security
 - **Disadvantages of LANs:**
 - LAN can be used Limited Area
 - Adds Complexity

- b. **WAN (Wide Area Network):** It connect computers and smaller networks to large networks over greater geographic areas, including different continents. These computers may be linked by means of cables, optical fibers or satellites. The purpose of WAN was to connect multiple systems in an efficient and economical way so that it can be shared by a large number of users.
- **Characteristics of WANs:**
 - Geographically Expansive
 - Centralizing Information
 - Remote Access
 - **Functions of WAN:**
 - Remote data
 - Facilitating Communication
- c. **MAN (Metropolitan Area Network):** A Network that expands into a metropolitan area and exhibits high data rates, high reliability and low data loss is called a metropolitan are network (MAN).

Characteristics of MANs:

- Intermediate Size
- High Speed Network
- Interconnects LANs
- Connects Whole City

NETWORKING DEVICES

Networking devices are hardware components or devices that are used to facilitate communication and data exchange between different computers and devices in a network. These devices play a crucial role in establishing and maintaining network connections, enabling data transmission, and managing network traffic. Here are some common types of networking devices:

1.Modem (Modulator/Demodulator)

Modem is a device that converts digital and analog signals At the source, modems convert digital signals to a form suitable for transmission over analog communication facilities (public telephone lines). At the destination, modems convert the signal back to a digital format.

2.Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they not only amplify the signal but also regenerate it. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.

3.Hubs- A hub (concentrator) contains multiple ports which is used to connect devices in a star topology. When a packet arrives at one port, it is copied to all the ports of the hub. But when the packets are copied, the destination address in the frame does not change to a broadcast address. It does this in a rudimentary way; it simply copies the data to all of the Nodes connected to the hub (broadcast).

4. Bridges- The main network device found at the data link layer is a bridge. This device works at a higher layer than the repeater and therefore is a more complex device. It has some understanding of the data it receives and can make a decision based on the frames it receives as to whether it needs to let the information pass, or can remove the information from the network. This means that the amount of traffic on the medium can be reduced and therefore, the usable bandwidth can be increased. Bridges are store and forward devices to provide error detection. They capture an entire frame before deciding whether to filter or forward the frame, which provides a high level of error detection because a frame's CRC checksum can be calculated by the bridge. Bridges are highly susceptible to broadcast storms. A broadcast storm occurs when several broadcasts are transmitted at the same time. It can take up huge bandwidth.

Advantages and Disadvantages of Bridges

Following are some advantages and disadvantages of Bridges:

Advantages:

- Reliability
- Manageability
- Scalability

Disadvantages:

- A bridge cannot filter out broadcast traffic.
- It introduces 20 to 30 % latency.
- Only 2 networks can be linked with a bridge

5.Router- Router is a networking device which forwards data packets along networks by using headers and forwarding/routing tables to determine the best path to forward the packets. Routers work at the Internet layer of the TCP/IP model or layer 3 of the OSI model. Routers also provide interconnectivity between like and unlike media. A router is connected to at least two networks, commonly two LANs or WANs or a LAN and its ISP's network. Some DSL and cable modems, for home use, have been integrated with routers to allow multiple home computers to access the Internet.

6. Switch: Switches are used to connect devices within a local network (such as a home or office network). They operate at the data link layer (Layer 2) of the OSI model and use MAC addresses to forward data only to the intended recipient device, increasing efficiency compared to older technologies like hubs.

Types of Switch

1. **Smart switches:** These switches have features similar to managed switches but are typically easier to set up and manage. They are suitable for small- to medium-sized networks.

2. **Layer 2 switches:** These switches operate at the Data Link layer of the OSI model and are responsible for forwarding data between devices on the same network segment.
 3. **Layer 3 switches:** These switches operate at the Network layer of the OSI model and can route data between different network segments. They are more advanced than Layer 2 switches and are often used in larger, more complex networks.
- 6. Gateway** – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

Connection-Oriented Service:

- It means establishing a dedicated connection between communicating entities before any data can be exchanged.
- The Connection establishment may be at the physical or the logical level and involves some form of signaling.
- The Service users undergo the following sequence of operations:
 - Establish a connection
 - Use the connection i.e. exchange of user data
 - Release or clear the connection

Connectionless Service:

- A Connectionless service is that which does not require a connection to be established before the exchange of data.
- Information is transferred by using independent data units.
- An example of a connectionless service is the postal mail
- Not safe as compare to Connection-Oriented Service
- expensive
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Distributed System:

- A Distributed System consists of multiple autonomous computers that communicate through a network.
- Distributed Computing Interconnects one or more personal computers/computational entities.
- It allows various services like file sharing, hardware sharing or network sharing.
- The purpose of the distributed system is to co-ordinate the use of shared resources or provides communication services to the users.

Characteristics or Features of a Distributed System:

- The System may consist of different kinds of computers and networks.
- This system uses distributed processing, in which a task is divided among multiple computers.
- A Computer Program that runs in a distributed system is called Distributed Program
- Links and the system may change during the execution of a distributed program.
- Each node has its own central processing unit with its own memory and communication hardware.
- The nodes or computers can be of different sizes, makes and computing powers.

Advantages of Distributed System:

- Distributed Database
- Cost Efficiency
- Faster Problem Solving
- Encapsulation
- Shared Resources
- Response Time Improvement

Disadvantages of Distributed System:

- Hardware Problems
- Technical Problem
- Data Integrity and Security Issues
- Maintenance of Remote Sites

Network software -Network software refers to computer programs and applications that enable the management, monitoring, security, communication, and overall functionality of computer networks. This software plays a crucial role in configuring, maintaining, and optimizing network operations.

Functions of Network Software

- Helps to set up and install computer networks
- Enables users to have access to network resources in a seamless manner
- Allows administrations to add or remove users from the network
- Helps to define locations of data storage and allows users to access that data
- Helps administrators and security system to protect the network from data breaches, unauthorized access and attacks on a network
- Enables network virtualizations

Network Design Issues:

Identifying and Justifying a Network: In the Network Design, Identifying and Justifying a network is necessary. The networks are capable of handling various data types and support a wide and ever-growing range of applications.

Scope: The scope of a network is bounded on one side by the communication facilities offered by the common carriers and on the other side by the applications.

Manageability: A Network comprising of many different facilities and with complex and specialized control procedures may be cost effective, it may be difficult to own.

Scalability: Another significant issue in general network design is the scalability – the scalability of the network to meet future demands based on projected traffic volumes etc.

Link Topology: The topological optimization of a network involves selecting the specific links interconnecting nodes.

Centralized and Decentralized System:

- In a Centralized System, all processing is done at one central computer.
- In the early days of computer technology, this type of processing was justified because data processing personnel were in short supply, hardware and software were expensive, and only large organization could afford computers.
- The Main advantage of Centralized system is being able to exercise tight control on system operations and applications.
- The Main disadvantage of Centralized system is lack of responsiveness to user's needs, because the system and its users could be located far apart from each other.
- In a Decentralized System, each user, department or division has its own computer for performing processing tasks.
- A Decentralized processing system is certainly more responsive to users than a centralized processing system.
- Decentralized systems have some drawbacks, including lack of coordination among organizational units, the high cost of having many systems and duplication of efforts.

Web Based Model:

- For ensuing scalability, the world-wide web has adopted a stateless approach to client-server communication called the Web-based Model.
- In this model, each interaction between the client and the server is independent of the other interactions.
- No permanent connection is established between the client and the server and the server maintains no state information about the clients.

A Web-based Model implies the following factors:

- Changing patterns of network use and unpredictable demands for bandwidth.

- Demand for increased amount of bandwidth.
- Demand for guaranteed quality of service in terms of bandwidth and minimum delay.
- Lack of control by IT staff.

TCP/IP Reference Model:

- Transmission Control Protocol / Internet Protocol is the protocol that makes possible the communication between different types of machines on different types of networks.
- It was developed by the United States Development of Defence Advanced Research Projects Agency (ARPA).
- ARPA originally created TCP/IP to connect military networks together.
- It is the most widely used protocol for interconnecting computers and it is the protocol of the Internet.
- TCP/IP provide the end to end communication.
- The main objective of TCP/IP provide the communication at long distance.

Goals of TCP/IP Model:

- The network should connect multiple networks together.
- The connection should withstand till the source and destination machines are functioning.
- The architecture should be so flexible that it should be able to transfer data among different hardware or software platforms.

Layers of TCP/IP Model:

1. Application Layer:

This is the highest level layer in the TCP/IP Model which is on the top of the transport layer. Application layer provide the interface between others layers.

This layer includes all processes and services that use the transport layer to deliver data.

The Protocols contained in the layer are FTP, SMTP, HTTP, SNMP, DNS, POP etc.

2. Transport Layer: The transport layer is responsible for end-to-end communication and data transfer between devices. It ensures that data is properly delivered, error-free, and in the correct order. The two primary transport layer protocols in the TCP/IP model are:

- **Transmission Control Protocol (TCP):** Provides reliable, connection-oriented communication. It ensures data delivery by implementing mechanisms like acknowledgment, flow control, and error correction.
- **User Datagram Protocol (UDP):** Offers connectionless communication and is used for applications where speed is more important than reliability. It doesn't guarantee data delivery or order, making it suitable for applications like streaming and online gaming.

3. Internet layer

The "Internet Layer" of the TCP/IP model is responsible for addressing, routing, and fragmenting data packets to facilitate their movement across different networks.

The key protocol at this layer is the Internet Protocol (IP).

IP is responsible for providing logical addressing and routing functionality, allowing data to be transmitted from a source to a destination across different networks.

IP addresses uniquely identify devices on a network and enable the routing of data packets to their intended destinations.

There are two main versions of the Internet Protocol:

1. **IPv4 (Internet Protocol version 4):** IPv4 is the older and more widely used version of the Internet Protocol. It uses 32-bit addresses, represented in four sets of decimal numbers (e.g., 192.168.1.1). However, the availability of IPv4 addresses has become limited due to the rapid growth of the Internet and the increasing number of connected devices.

2. **IPv6 (Internet Protocol version 6):** IPv6 was introduced to address the limitations of IPv4 and to provide a solution to the impending exhaustion of IPv4 addresses. IPv6 uses 128-bit addresses, represented in eight sets of hexadecimal numbers)

Both IPv4 and IPv6 are integral to the functioning of the Internet Layer in the TCP/IP model. They allow for the identification of devices, as well as the routing and delivery of data packets across various networks to reach their intended destinations.

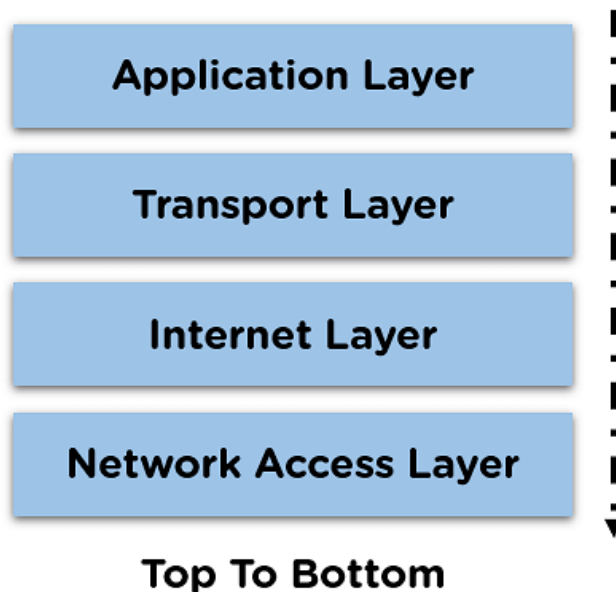
3. **ICMP:** - It stands for internet control message protocol. ICMP informs the host about the problem occurred in the network.

4. Network access layer: -

It is the lowest layer of TCP/IP layer.

It is the combination of physical layer and data link layers.

Frame relay, Ethernet, x.25 token ring protocol use in this layer



Peer-to-Peer Model:

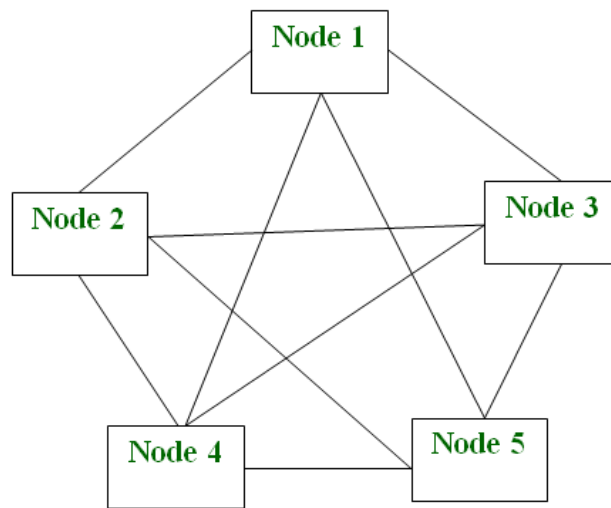
- It was invented in 1979
- Devices are connected directly which is use for share the data and file from one computer to other.
- Without central server user can send data, file form one computer to other computer that is also called point to point connection and p2p connection.
- Every connected computer work like client and server.
- With the help of internet, we can connect the device with other. there are no central devices in this model and so user can contact directly to other device for uploading and downloading the file.

Advantages of Peer to Peer Network Model:

- Useful for small business
- Cheaper because it does not have server.
- Manage, installation and setup easy.
- If one computer get damage, there will be no effect on other connected devices/network.
- There is no need of manager for managing the device because every computer manages their own devices.

Disadvantages of Peer to Peer Network Model:

- Not very secure so hacker can easily hack the devices.
- We cannot connect more than 10 computers.
- No central point of storage for file archiving.
- Virus can easily enter in computers.
- Requires users to administer their own computers.
- P2P do not have server so cannot get back up.



P2P Architecture

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Client/Server Network Model:

The client-server architecture is a model of computer networking in which a client requests services hosted by a server. In this model, the client sends a request to the server for a particular service, and the server accepts this request and provides the service to the client.

- **In** this model multiple devices connected by the central device
- In Client-Server Network, certain computers act as server and other act as clients.
- A server is the computer that makes the network resources available and provides services to other computers on their request.
- A Client is any computer on the network that requests something from a server.
- Also called request- response model.

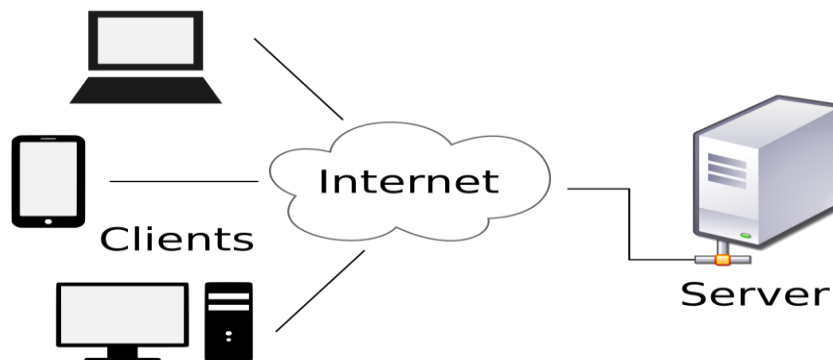
Advantages of Client/Server Model:

- Centralized data storage allows users to access information and provides easy back up of critical data.
- Easy manageability of a large number of users.
- Central security is provided.
- We can connect large amount of computer.
- Ability to share expensive equipment such as laser printers.

- It can use for small and big network.
- We can share the file easily.

Disadvantages of Client/Server Model:

- If server get down its will effect to whole connected computers.
- We need different technique to maintain a network.
- Complex installation and configuration as compared to a peer to peer network.
- Requirement of client computers and high speed server computers with lots of memory and disk space.



FRAME RELAY: -

- It is protocol which is used in wide area network.
- It is packet switching technology that is used to transfer the data between two and multiples local area networks.
- It is designed to transfer the data between local area network at the low cost.
- Frame relay, OSI model used in physical layer and data link layer.
- The speed of data transfer of frame relay is 65 kbps to 45 mbps.
- Frame relay use because when sender send the data in packet the speed of data transmission gets slow. when put the data in frame and send over the network to end of destination point.
- So it increases the speed of data transmission.
- Frame relay use virtual circuit to connect single router to multiple computers.
- **There are two types of frame relay**

- 1. **PVC:** Permanent virtual circuit. PVC is the permanent connection between frame relay node which can be used for long time. These connections are static and cannot change with time
- 2. **SVC:** Switch virtual circuit. SVC is the temporary connection between frame relay node which can be used for used time.

ADVANTAGE OF FRAME REALY: -

- 1. It provide the high speed because error is in this protocol that why less overhead.
- 2. lost cost by using frame relay we can transfer the data at low cost.
- 3. It provide frame size 9000 bytes which is enough for LAN frame types.
- 4.IT provide safe connection.
- 5.Less delay

Disadvantage of frame relay: -

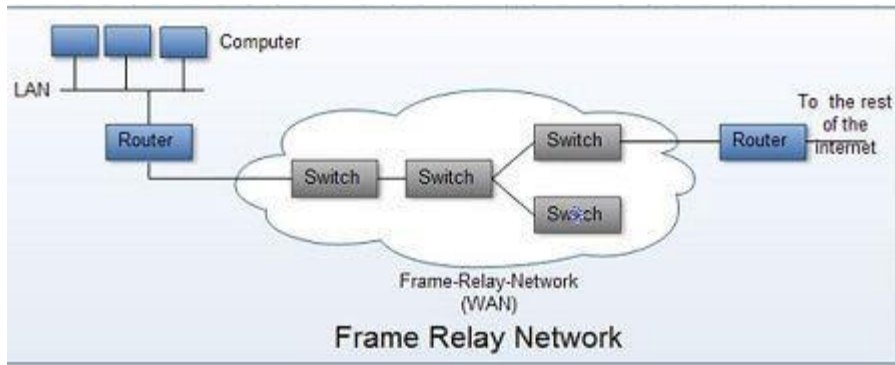
- Cannot flow control and error control.
- We did not get the data in correct sequence.
- It provides the data transfer speed 64 kb to 45 MB. It is not suitable for the

High data rates.

- It is not suitable for sending sensitive data (real time audio, video,).

Characteristics of frame relay: -

- Frame relay work only physical layer and data link layer.
- It is connection less services.
- It is provided the data speed of 64 KB to 45 MB.
- It can detect the error but cannot control the error.



X.25 MODEL

- X.25 is a protocol that was developed by the Telecommunication Standardization sector (ITU-T) Of international telecommunication union.
- It is packet switch communication over WAN.
- It is used in ATM and credit card verification.
- This protocol **is** based on the protocols used **in** early packet switching networks such **as** ARPANET, DATAPAC, and TRANSPAC.
- X.25 Packet Switched networks allows remote devices to communicate with each other across high speed digital links.
- X.25 is a Telecommunications Standardization Sector (ITU-T) recommendation that defines the physical level (physical layer), link level (data link layer), and packet level (network layer) of the open systems interconnection (OSI) reference model.
- It is a connection oriented services.
- X.25 is a connection oriented service. It supports switched virtual circuits as well as the permanent circuits
- A protocol X.21 which **is** a physical layer protocol **is** used to specify the physical electrical and procedural **interface** between the host and network.

ATM Architecture:

The Physical Layer:

- **Physical Layer** – This layer corresponds to physical layer of OSI model. At this layer, the cells are converted into bit streams and transmitted over the physical medium. This layer has two sub layers: PMD sub layer (Physical Medium Dependent) and TC (Transmission Convergence) sub layer.
- **The ATM Layer:**

ATM Layer –This layer is comparable to data link layer of OSI model. It accepts the 48 byte segments from the upper layer, adds a 5 byte header to each segment and converts into 53 byte cells. This layer is responsible for routing of each cell, traffic management, multiplexing and switching

This layer is common to all the services that provide the packet transfer capabilities.

The ATM layer deals with cells and transport of cell.

It also deals with establishment and release of virtual circuits.

ATM Adaptation Layer:

- This layer corresponds to network layer of OSI model. It provides facilities to the existing packet switched networks to connect to ATM network and use its services. It accepts the data and converts them into fixed sized segments. The transmissions can be of fixed or variable data rate. This layer has two sub layers – Convergence sub layer and Segmentation and Reassembly sub layer.

Advantages of ATM:

- It provides flexibility for different applications.
- It has ability to connect LAN to WAN.
- It is best suited for voice, video and data communication.
- It provides scalability in speed and network size.
- It transmits data at high speed mbps.

Disadvantages of ATM:

- It is an expensive technique.
- The mechanisms are complex for achieving Quality of Service (QoS).

ATM Applications:

- Video Conferencing.
- Distance Learning

OSI MODELS: -

The full name of the OSI model is "Open System Interconnection." It was developed by the International Organization for Standardization (ISO) in 1984. This model consists of 7 layers.

The OSI model serves as a reference model for communication between two users in a network. Each layer of this model is not dependent on the other layers, but data transmission occurs from one layer to another.

The OSI model explains how data or information is sent and received in a network. Each layer in this model has its specific role, ensuring that data can easily travel from one system to another.

The OSI model also describes how network hardware and software interact with each other in terms of layers.

This model consists of 7 layers, and each layer has its distinct function.

1.Physical Layer: -

The Physical Layer is the lowest layer in the OSI model.

This layer is responsible for physical and electrical connections, such as voltage levels, data rates, etc.

In this layer, digital signals are converted into electrical signals. The layout of the network, also known as the network's topology, is defined in this layer.

The Physical Layer also determines whether communication will be wireless or wired.

It is also referred to as the "**bit unit.**"

Tasks of the Physical Layer:

1. **Physical Connection:** The Physical Layer explains how two or more devices are physically connected to each other.
2. **Transmission Mode:** It defines the transmission mode used between two devices in the network. There are three transmission modes: simplex, half-duplex, and full duplex.
3. **Signal Specification:** It determines the characteristics of the signals used to transmit information.
4. **Network Topology:** The Physical Layer is responsible for managing the network topology.

2.Data Link Layer

The Data Link Layer is the second layer from the bottom in the OSI model.

This layer is also referred to as the "**Frame Unit.**"

In the Data Link Layer, packets of data sent by the Network Layer are encoded and decoded. Additionally, this layer ensures that there are no errors in the data packets.

Tasks of the Data Link Layer:

1. **Framing:** The Data Link Layer takes the packets from the Network Layer and encapsulates them into frames. These frames consist of a header and a trailer, which contain control information needed for data transmission and error detection.
2. **Physical Addressing:** It provides physical addressing (such as MAC addresses) to devices on the network. MAC addresses are unique identifiers assigned to network interfaces.

3. **Error Detection and Correction:** The Data Link Layer is responsible for detecting and, in some cases, correcting errors that might occur during transmission. This is done using techniques like checksums or cyclic redundancy checks (CRC).
4. **Flow Control:** The Data Link Layer helps in managing the flow of data between devices with different data transfer speeds. It prevents fast senders from overwhelming slow receivers by implementing flow control mechanisms.
5. **Access Control:** In shared network mediums like Ethernet, the Data Link Layer helps manage access to the medium, ensuring that multiple devices can communicate without causing collisions.

3.NETWORK LAYER

The Network Layer is the third layer in the OSI model and is also referred to as the "**Packet Unit.**".

The primary role of this layer is to provide logical addresses, such as IP addresses, to devices.

The data in the Network Layer is in the form of data packets, and it is responsible for delivering these data packets from one device to another.

Tasks of the Network Layer:

- 1 **IP Addressing:** The main function is to assign IP addresses to devices.
- 2 **Packet Routing:** The Network Layer determines the optimal path for data packets to travel from the source to the destination across the network. It selects the best route based on factors like distance, congestion, and cost.
- 3 **Addressing and Forwarding:** It adds source and destination addresses to the header of data packets. These addresses are used to identify devices on the Internet.
- 4 **Routing:** The Network Layer decides the best path for data packets to follow based on routing algorithms, ensuring efficient and reliable data delivery.
- 5 **Switching:** The Network Layer also performs switching, which involves the forwarding of data packets within a local network (like a LAN) based on MAC addresses.

4.TRANSPORT LAYER

The Transport Layer is the fourth layer in the OSI model, and it is also known as the "**Segment Unit.**"

This layer is responsible for transferring data between network entities in an accurate and reliable manner. It ensures that data is transmitted without errors.

The Transport Layer also verifies that the data sent in a particular order is received in the same order. Its role includes managing communication between two computers.

The Transport Layer is sometimes referred to as the "end-to-end" layer because it provides point-to-point connections for transferring data. It ensures that data is reliably transmitted from the source to the destination.

This layer offers two types of services: **connection-oriented and connectionless**.

In connection-oriented communication, a connection is established, data is transmitted, and then the connection is terminated. **In connectionless communication**, data packets are sent independently without establishing a formal connection.

The tasks of the Transport Layer are as follows:

1. **Data Transfer:** The primary function of the Transport Layer is to transfer data from one computer to another.
2. **Facilitates Communication:** It provides communication capabilities between two computers, enabling them to exchange data.
3. **Point-to-Point Connection:** The Transport Layer establishes point-to-point connections for data transfer.
4. **Segmentation:** When this layer receives a message from the upper layers, it divides the message into multiple segments. Each segment is assigned a sequence number for easy identification.
5. **Flow Control and Error Control:** The Transport Layer manages both flow control, which ensures that data is sent at a rate the receiver can handle, and error control, which involves detecting and correcting errors that may occur during transmission.

5.Session Layer: - The Session Layer is the fifth layer in the OSI model, and it manages connections between multiple computers.

The Session Layer provides sessions for communication between two devices. In simple terms, whenever a user opens a website, a session is established between the user's computer and the website's server.

the primary function of the Session Layer is to handle the establishment, maintenance, and termination of connections.

Tasks of the Session Layer:

1. **Session Establishment, Maintenance, and Termination:** The main role of the Session Layer is to establish, maintain, and terminate sessions between two devices.
2. **Dialog Controller:** Similar to a dialog controller, the Session Layer manages the communication between two processes. It creates a dialog between processes on different devices.
3. **Synchronization:** The Session Layer ensures synchronization between sender and receiver by coordinating the flow of data and detecting errors or lost data. If an error occurs, the transmission is retransmitted.

6.Presentation Layer

The Presentation Layer is the sixth layer in the OSI model.

This layer is responsible for tasks like data encryption and decryption.

It is also used for data compression. The Presentation Layer is closely related to the operating system.

The Presentation Layer is sometimes referred to as the "**syntax layer**" because it maintains the proper syntax of data.

Functions of the Presentation Layer:

1. **Encryption and Decryption:** The primary function of this layer is to perform encryption and decryption. Encryption helps in securing data. Through encryption, data can be kept confidential and protected from unauthorized access.
2. **Data Compression:** The Presentation Layer also handles data compression. Compression is crucial as it reduces the size of data, making it more efficient to transmit and store.

In summary, the Presentation Layer handles tasks related to data encryption, decryption, compression, and translation. It ensures that data is presented in the correct format for communication and handles security and efficiency aspects of data transmission.

7.Application Layer

The Application Layer is the seventh (and topmost) layer in the OSI model.

Its primary function is to provide an interface between our actual applications and the other layers of the OSI model.

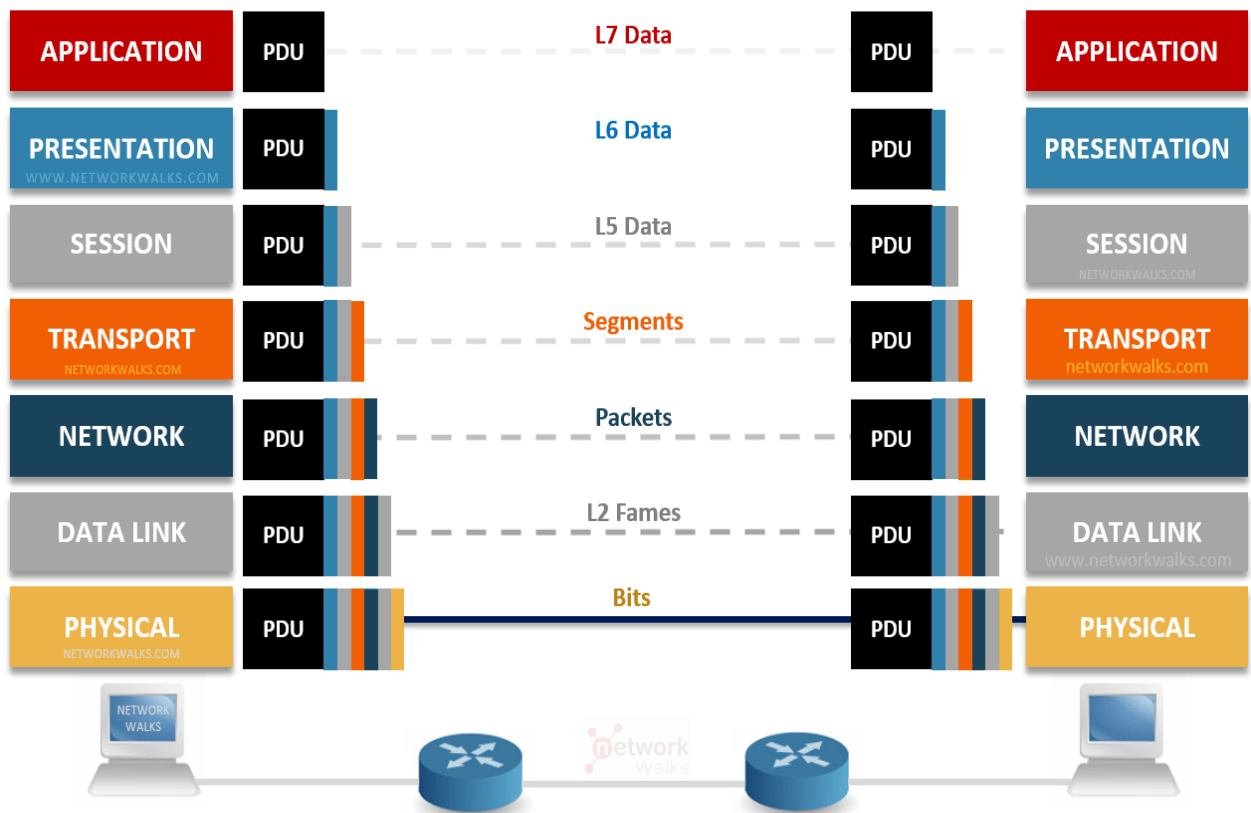
The Application Layer is the closest layer to the end user, providing network services to end users. It's responsible for presenting data to the user and enabling communication between different applications.

Protocols like HTTP, FTP, SMTP, and POP fall under the scope of this layer. It controls how applications access the network.

Functions of the Application Layer:

1. **File Access and Management:** The Application Layer allows users to access and retrieve files from other devices on the network.
2. **Email Services:** It enables email forwarding and storage, allowing users to send, receive, and manage emails.
3. **Database Access:** Users can access databases and directories using the Application Layer.

In summary, the Application Layer provides the interface for end users to interact with network services. It facilitates tasks such as accessing files, managing emails, and interacting with databases. This layer acts as the bridge between the user and the underlying network infrastructure.



Let's see the differences between the OSI and TCP/IP model in a tabular for:

OSI Model	TCP/IP Model
It stands for Open System Interconnection .	It stands for Transmission Control Protocol .
OSI model has been developed by ISO (International Standard Organization).	It was developed by ARPANET (Advanced Research Project Agency Network).
In the OSI model, the transport layer provides a guarantee for the delivery of the packets.	The transport layer does not provide the surety for the delivery of packets. But still, we can say that it is a reliable model.
This model is based on a vertical approach.	This model is based on a horizontal approach.
In this model, the session and presentation layers are separated, i.e., both the layers are different.	In this model, the session and presentation layer are not different layers. Both layers are included in the application layer.
It is also known as a reference model through which various networks are built. For example, the TCP/IP model is built from the OSI model. It is also referred to as a guidance tool.	It is an implemented model of an OSI model.
In this model, the network layer provides both connection-oriented and connectionless service.	The network layer provides only connectionless service.
Protocols in the OSI model are hidden and can be easily replaced when the technology changes.	In this model, the protocol cannot be easily replaced.
It consists of 7 layers.	It consists of 4 layers.
OSI model defines the services, protocols, and interfaces as well as provides a proper distinction between them. It is protocol independent.	In the TCP/IP model, services, protocols, and interfaces are not properly separated. It is protocol dependent.
The usage of this model is very low.	This model is highly used.
It provides standardization to the devices like router, motherboard, switches, and other hardware devices.	It does not provide the standardization to the devices. It provides a connection between various computers.

